Chapter 2 Mathematical formulation and stopping theory

2.1 Probability model and admissible information

Fix a finite horizon N and a deterministic earliest permitted stop $m \in \{0, \dots, N\}$. Work on a probability space $(\Omega, \mathcal{A}, \mathbb{P})$. Let Y^* be the reference answer, H_t the information actually available after decision step t, and

$$\mathcal{F}_t = \sigma(H_0, \dots, H_t), \qquad A_t = a_t(H_0, \dots, H_t), \qquad C_t = \mathbf{1}\{A_t = Y^*\}.$$

For empirical evaluation, correctness is the versioned domain grader $C_t = g_d(A_t, Y^*) \in \{0, 1\}$. Exact answer equality is the special case displayed above. The binary-reward arguments remain unchanged for this grading predicate. This notation identifies the measured endpoint; it does not certify semantic correctness of every stored label. Appendix E illustrates the information restrictions and delayed-repair counterexample.

All observations, executed peer calls, verifier outputs, and controller randomness used by a decision belong in this filtration. The runtime interface does not supply offline ground-truth fields, future revisions, future peer responses, or features computed using an unfinished trace. These are interface restrictions; mathematical measurability is governed by the information model and the caveat below. The correctness indicator need not be observable, even though the answer is.

The filtration is a statistical information model, not a model of computational hardness. If the reference answer is a known deterministic measurable function of a fully revealed task, then the selected candidate's correctness is $\mathcal{F}_t$-measurable and $q_t = C_t$, even

without a gold input field. All results below still hold in that degenerate case. Nondegenerate conditional uncertainty requires a specified latent-reference generative model or an explicitly coarser filtration of decision statistics, with admissible policies restricted to that information. A fitted probability based on prefix summaries does not establish a posterior conditioned on all mathematical semantics of a task.

The candidate A_t means the answer that the specified policy would emit if it stopped at t. If the policy selects among current, earlier, or peer answers, define A_t by that causal selection procedure. Results for the current generator answer do not automatically apply to a different selected answer. If every available candidate in a finite, observable candidate set can be selected freely, the optimal immediate correctness reward is $\max_{a \in \mathcal{B}_t} \mathbb{P}(Y^* = a \mid \mathcal{F}_t)$, where $\mathcal{B}_t$ is the candidate set available at t. With the empirical grading predicate, the analogous immediate reward is $\max_{a \in \mathcal{B}_t} \mathbb{E}[g_d(a, Y^*) \mid \mathcal{F}_t]$.

A policy's stop time τ is admissible when

$$m \le \tau \le N, \qquad \{\tau \le t\} \in \mathcal{F}_t \quad \text{for every } t.$$

Let K_t be the nonnegative, nondecreasing, integrable, adapted cumulative cost of **all** computation executed by t, including peers and probes. Put

$$q_t = \mathbb{E}[C_t \mid \mathcal{F}_t], \qquad G_t = q_t - K_t, \qquad c_t = \mathbb{E}[K_{t+1} - K_t \mid \mathcal{F}_t].$$

The objective is $\sup_\tau \mathbb{E}[C_\tau - K_\tau]$. The step-cost specialization is $K_t = \lambda(t - m)$ for $t \ge m$; previously incurred constant costs do not change the optimizer. The empirical choice $\lambda = 0.05$ is a utility convention, not a universal monetary cost or an assertion about token savings.

Lemma 1 (observable reward reduction). For every admissible stop time,

$$\mathbb{E}[C_\tau - K_\tau] = \mathbb{E}[G_\tau].$$

Proof. The event $\{\tau = t\}$ belongs to $\mathcal{F}_t$, so $\mathbb{E}[\mathbf{1}_{\{\tau=t\}} C_t] = \mathbb{E}[\mathbf{1}_{\{\tau=t\}} q_t]$. Sum this equality over the finite set $t = m, \dots, N$, and subtract $\mathbb{E}[K_\tau]$. □

For stopping-only comparisons, a full potential trace may be defined even after a policy would stop. Its prefixes must have the same law as the actual generation process up to stopping. A controller that changes prompts, sampling, peer allocation, or answer selection changes the process; a retrospective evaluation must account for that change.

2.2 Exact binary correctness drift

For $t < N$, define the conditional joint transition masses

$$r_t = \mathbb{P}(C_t = 0, C_{t+1} = 1 \mid \mathcal{F}_t), \qquad s_t = \mathbb{P}(C_t = 1, C_{t+1} = 0 \mid \mathcal{F}_t),$$

and the conditional repair and corruption probabilities

$$\alpha_t = \begin{cases} r_t/(1-q_t), & q_t < 1, \\ 0, & q_t = 1, \end{cases} \qquad \beta_t = \begin{cases} s_t/q_t, & q_t > 0, \\ 0, & q_t = 0. \end{cases}$$

The value assigned on a zero-probability conditioning state is arbitrary in $[0, 1]$: its multiplier is zero. These quantities are probabilities per decision transition, not continuous-time intensities. No Markov, independence, monotonicity, or observability of C_t is needed for the following identity.

Theorem 2 (conditional drift identity). Almost surely,

$$\mu_t := \mathbb{E}[G_{t+1} - G_t \mid \mathcal{F}_t] = (1 - q_t)\alpha_t - q_t\beta_t - c_t.$$

Proof. Pointwise, $C_{t+1} - C_t = \mathbf{1}_{\{C_t=0, C_{t+1}=1\}} - \mathbf{1}_{\{C_t=1, C_{t+1}=0\}}$. The tower property and nested filtrations give

$$\mathbb{E}[q_{t+1} \mid \mathcal{F}_t] = \mathbb{E}[\mathbb{E}[C_{t+1} \mid \mathcal{F}_{t+1}] \mid \mathcal{F}_t] = \mathbb{E}[C_{t+1} \mid \mathcal{F}_t].$$

Subtract $q_t = \mathbb{E}[C_t \mid \mathcal{F}_t]$, use the indicator identity, and subtract c_t. □

Consequently, with $D_m = 0$ and $D_t = \sum_{s=m}^{t-1} \mu_s$, $M_t = G_t - G_m - D_t$ is an integrable martingale. This is a discrete Doob decomposition. The process q_t need not be a martingale: it concerns the changing answer C_t, rather than the conditional probability of one fixed event.

For a sample of n observed transitions, let n_0, n_1 be the numbers of current incorrect and correct states and let n_{01}, n_{10} count repairs and corruptions. On that same sample,

$$\left(1 - \frac{n_1}{n}\right) \frac{n_{01}}{n_0} - \frac{n_1}{n} \frac{n_{10}}{n_1} = \frac{n_{01} - n_{10}}{n} = \frac{1}{n} \sum_i (C_{i,t+1} - C_{i,t}),$$

with the zero-denominator convention above. Frequencies n_{01}/n and n_{10}/n are already joint event frequencies; multiplying them again by $1 - q$ and q is incorrect. Matching this sample identity verifies arithmetic, not an online probability model. A product of separately averaged, instance-specific probabilities generally differs from the average of their products.

Corollary 2.1 (affine stakes). Suppose a correct answer earns $v \geq 0$ and an incorrect answer loses $p \geq 0$, with $v + p > 0$, and computation still costs K_t. The observable stop reward and its conditional drift become

$$G_t^{v,p} = v q_t - p(1 - q_t) - K_t = (v + p) q_t - p - K_t,$$

$$\mu_t^{v,p} = (v + p)\big((1 - q_t)\alpha_t - q_t \beta_t\big) - c_t.$$

Proof. The realized terminal reward is $vC_t - p(1 - C_t) - K_t$. Apply Lemma 1 and the linearity of conditional expectation to Theorem 2. □

For constant computation cost, the correctness-versus-compute trade-off is therefore determined by cost divided by $v + p$; $-p$ is an additive constant. Increasing stakes can change the optimal policy. Under a fixed potential trace law and nondecreasing computation costs, optimal continuation regions expand as $v + p$ increases; Corollary 3.1 proves the corresponding ordering of earliest optimal stop times. That ordering need not hold for empirically thresholded policies, or when stakes change the generation process, answer selection, or costs. All comparisons must state both scales.

2.3 General finite-horizon optimal stopping

Theorem 3 (Bellman recursion and Snell envelope). Set

$$S_N = G_N, \qquad S_t = \max\{G_t, \mathbb{E}[S_{t+1} \mid \mathcal{F}_t]\}, \quad t = N-1, \dots, m.$$

Then S is the smallest integrable supermartingale dominating G. For each $t \geq m$,

$$S_t = \operatorname*{ess\,sup}_{\tau:\, t \leq \tau \leq N} \mathbb{E}[G_\tau \mid \mathcal{F}_t], \qquad \tau_t^* = \inf\{s \in \{t, \dots, N\} : S_s = G_s\}$$

is an optimal stopping time, and $S_t = \mathbb{E}[G_{\tau_t^*} \mid \mathcal{F}_t]$.

Proof. Backward induction gives integrability, adaptation, $S_t \geq G_t$, and $S_t \geq \mathbb{E}[S_{t+1} \mid \mathcal{F}_t]$. If W is any integrable supermartingale with $W_s \geq G_s$, then $W_N \geq S_N$. If $W_{s+1} \geq S_{s+1}$, the supermartingale property and domination give $W_s \geq \max\{G_s, \mathbb{E}[S_{s+1} \mid \mathcal{F}_s]\} = S_s$. This proves minimality by induction.

For any bounded admissible τ, finite optional sampling, or the finite telescoping sum of conditional supermartingale increments, gives $\mathbb{E}[G_\tau \mid \mathcal{F}_t] \le \mathbb{E}[S_\tau \mid \mathcal{F}_t] \le S_t$. The first contact time is measurable and no later than N. On $\{s < \tau_t^*\}$, $S_s > G_s$, so the recursion forces $S_s = \mathbb{E}[S_{s+1} \mid \mathcal{F}_s]$. Thus $(S_{s\wedge\tau_t^*})_{s=t}^N$ is a martingale. At contact $S_{\tau_t^*} = G_{\tau_t^*}$, giving equality in the upper bound. Attainment establishes the essential-supremum representation. □

This is the standard finite-horizon optimal stopping construction; see Ferguson, Chapter 3, Section 3.2. The proof here specializes it to the observable correctness-minus-cost reward and includes the explicit protocol floor.

The optimal continuation advantage is

$$\Delta_t^* = \mathbb{E}[S_{t+1} \mid \mathcal{F}_t] - G_t = \mu_t + \mathbb{E}[S_{t+1} - G_{t+1} \mid \mathcal{F}_t] \ge \mu_t.$$

The second term is the value of future choices. Therefore $\mu_t > 0$ is sufficient to favor continuation at $t < N$; $\mu_t \le 0$ alone is insufficient to justify stopping. At $N - 1$ the terms coincide. The optimal policy stops at the first $\Delta_t^* \le 0$, or at N.

Corollary 3.1 (stake monotonicity under a fixed process). Compare affine stakes with $0 < w_1 \le w_2$, where $w_j = v_j + p_j$. Suppose that the filtration, candidate answers, potential trace law, horizon, floor, and nondecreasing cumulative cost process are identical in both problems. Then their earliest optimal contact times satisfy $\tau^*(w_1) \le \tau^*(w_2)$ almost surely.

Proof. For $t < N$, the normalized advantage of being required to continue at least once is

$$\frac{\Delta_t^*(w)}{w} = \operatorname*{ess\,sup}_{\tau:\, t+1 \le \tau \le N} \mathbb{E}\left[q_\tau - q_t - \frac{K_\tau - K_t}{w} \,\middle|\, \mathcal{F}_t \right].$$

The constant incorrect-answer offset cancels. Since $K_\tau - K_t \geq 0$, each conditional expectation is nondecreasing in w, as is its essential supremum. Consequently $\Delta_t^*(w_1) > 0$ implies $\Delta_t^*(w_2) > 0$. Along any fixed path, the larger-stakes policy cannot stop while the smaller-stakes policy has continued at every decision so far: each of those strict continuation conditions also holds for larger stakes. With the same tie convention and forced terminal stop, this proves the stated ordering. □

This result does not assert that accuracy itself increases strictly, that all fitted drift boundaries obey the ordering, or that a negative drift becomes positive merely because stakes increase.

2.4 When a drift-sign boundary is optimal

Theorem 4 (persistent nonpositive drift). Define

$$T_c = \min\big(\{t \in \{m, \dots, N-1\} : \mu_t \leq 0\} \cup \{N\}\big).$$

Assume, almost surely, that $\mu_s \leq 0$ for every $s \in \{T_c, \dots, N-1\}$. Then T_c maximizes $\mathbb{E}[G_\tau]$ over all admissible stop times.

Proof. For any such τ, telescoping and the measurability of $\{\tau > s\}$ yield

$$\mathbb{E}[G_\tau] = \mathbb{E}[G_m] + \mathbb{E}\sum_{s=m}^{N-1} \mathbf{1}_{\{\tau > s\}} \mu_s.$$

For $s < T_c$, $\mu_s > 0$ and $\mathbf{1}_{\{T_c > s\}} = 1$. For $s \geq T_c$, $\mu_s \leq 0$ and $\mathbf{1}_{\{T_c > s\}} = 0$. Accordingly,

$$(\mathbf{1}_{\{T_c > s\}} - \mathbf{1}_{\{\tau > s\}})\mu_s \geq 0$$

on every path. Sum, take expectations, and apply the telescoping identity to T_c. □

This is a condition on the **conditional drifts along sample paths**. A single crossing of an empirical population mean curve does not establish it. Theorem 4 is a special case of monotone stopping problems, whose one-stage look-ahead conditions are discussed in Ferguson, Chapter 5.

Corollary 4.1 (a sufficient structural condition). Suppose cost is constant $c_t = \lambda$, and, almost surely along every path, $q_{t+1} \geq q_t$, $\alpha_{t+1} \leq \alpha_t$, and $\beta_{t+1} \geq \beta_t$ for $m \leq t < N-1$, for specified versions of the hazards. Then μ_t is nonincreasing and Theorem 4 applies.

Proof. Subtract adjacent drift expressions:

$$\begin{aligned}\mu_{t+1} - \mu_t = (1-q_{t+1})(\alpha_{t+1}-\alpha_t) + \alpha_t(q_t - q_{t+1}) \\ - q_{t+1}(\beta_{t+1}-\beta_t) - \beta_t(q_{t+1}-q_t).\end{aligned}$$

Every term is nonpositive because all probabilities lie in $[0,1]$. With variable predictable cost an extra term $-(c_{t+1}-c_t)$ appears; nondecreasing c_t is an additional sufficient condition. □

These sufficient assumptions are strong. They are not established by a high correctness AUC, a good average utility curve, or a monotone fitted hazard curve. A noisily estimated posterior may increase and decrease after new observations even when the marginal accuracy improves.

Exact counterexample 1: delayed repair

Use times $0, 1, 2$, constant cost $1/10$, and correctness probabilities $q = (0, 0, 1)$. An exact-answer realization has a latent binary reference Y^*, a surely incorrect placeholder at times 0 and 1, and a generator that outputs Y^* at time 2. No information at time 0 or 1 reveals which reference answer will be returned. Then $G = (0, -1/10, 4/5)$ and

$\mu_0 = -1/10$. The myopic boundary stops at 0 with reward 0. The optimal stop is 2 with reward $4/5$. A second system with $q = (0, 0, 0)$ has the same current $(q_0, \alpha_0, \beta_0, c_0) = (0, 0, 0, 1/10)$ but optimally stops at 0. Thus even the exact current hazard triple cannot identify a general multi-step optimal policy. The example can be shifted to a floor $m = 2$ without changing the comparison.

Exact counterexample 2: a population curve misses adaptive value

Take $q_0 = 1/2$ and cost $1/10$. At time 1, a fair observed signal gives branch A with $q_1 = 2/5$ or branch B with $q_1 = 3/5$. At time 2, set $q_2 = 1$ on A and $q_2 = 0$ on B. Both unconditional one-step drifts equal $-1/10$, and the deterministic-time expected rewards are $1/2, 2/5, 3/10$. Yet continuing once, then continuing on A and stopping on B, achieves

$$\tfrac{1}{2}(1 - 2/10) + \tfrac{1}{2}(3/5 - 1/10) = 13/20 > 1/2.$$

For an exact-answer realization, let $Y^* \in \{0, 1\}$ have prior probability $1/2$ of 1, and use candidate answer 1 at times 0 and 1. Set $\mathbb{P}(A, Y^* = 1) = 1/5$, $\mathbb{P}(A, Y^* = 0) = 3/10$, $\mathbb{P}(B, Y^* = 1) = 3/10$, and $\mathbb{P}(B, Y^* = 0) = 1/5$. At time 2 the generator returns Y^* on branch A and a surely incorrect placeholder on branch B. The controller uses only the observed branch to make its time-1 decision. Thus the examples respect a common reference answer and use no hindsight information to make decisions.

2.5 Partial observation and feature models

For a causal feature vector $Z_t = \phi_t(H_0, \dots, H_t)$, define

$$\begin{aligned}
\tilde{q}_t &= \mathbb{P}(C_t = 1 \mid Z_t), \\
\widetilde{\alpha}_t &= \mathbb{P}(C_{t+1} = 1 \mid C_t = 0, Z_t), \\
\widetilde{\beta}_t &= \mathbb{P}(C_{t+1} = 0 \mid C_t = 1, Z_t).
\end{aligned}$$

Their hazard expression equals $\mathbb{E}[C_{t+1} - C_t \mid Z_t]$ exactly, subject to the same null-state convention. This feature-conditional quantity need not equal the history-conditional drift in Theorem 2. In particular, unrelated $\sigma(Z_t)$ at different times need not form a filtration.

A Bellman equation on a compressed state requires sufficiency: conditional on that state, the law of future observations, reward, and cost must not depend on omitted history. Alternatively a correctly specified partially observed Markov model can use the full posterior over its latent state as the belief state. A scalar correctness posterior and two current hazards do not by themselves provide that transition model. Without such assumptions, a fitted hazard rule is an empirical myopic approximation.

Retrospective correctness classifiers, continuation predictors, and optimal stopping policies are different estimands. Training with labels is allowed; using those labels or future observations to decide on the same live instance is not. A prefix-stable feature transform must be fitted before evaluation or updated solely from information available by that prefix. Full-family or full-trace normalizations may change earlier decisions when future rows are added, even without directly using correctness labels.

2.6 Calibration and perturbation bounds

Discrimination is not probability accuracy. AUC is unchanged by a strictly increasing transform, while the drift sign generally changes. Even marginal calibration $\mathbb{E}[C_t \mid \hat{q}_t] = \hat{q}_t$ does not imply $\hat{q}_t = \mathbb{E}[C_t \mid \mathcal{F}_t]$: a constant $1/2$ score is marginally calibrated in a balanced population whose available history perfectly separates correct and incorrect cases.

Proposition 5 (class weighting changes the population target). For a binary target with conditional probability $p \in (0,1)$, positive class weights w_1, w_0, and an unconstrained probability prediction r, the weighted Bernoulli log loss is uniquely minimized at

$$r = \frac{w_1 p}{w_1 p + w_0(1-p)}, \qquad p = \frac{w_0 r}{w_1(1-r) + w_0 r}.$$

Proof. Differentiate $-w_1 p \log r - w_0(1-p)\log(1-r)$; setting the derivative to zero yields the first formula. Strict positivity of the second derivative gives uniqueness. Algebra gives the inverse. □

`equation_analysis.py` and `trace_analysis.py` fit class-balanced classifiers. Thus their uncorrected probability outputs are not guaranteed to be target-distribution q, α, β, even if ranking is useful. For example, balanced weighting in a population with prevalence $1/10$ maps its constant posterior to $1/2$. With $\alpha = 1/50, \beta = 1/200, \lambda = 1/100$, changing only q from $1/10$ to $1/2$ changes drift from $3/400 > 0$ to $-1/400 < 0$. The inverse formula describes a population optimum; model misspecification, regularization, estimated weights, and distribution shift still require independent probability validation. ECE and Brier scores are useful diagnostics but do not provide uniform conditional error guarantees.

Proposition 6 (drift perturbation and interval propagation). If $|\hat{q} - q| \le \varepsilon_q$, $|\widehat{\alpha} - \alpha| \le \varepsilon_\alpha$, $|\widehat{\beta} - \beta| \le \varepsilon_\beta$, and $|\hat{c} - c| \le \varepsilon_c$, with all probabilities in $[0,1]$, then

$$|\widehat{\mu} - \mu| \le 2\varepsilon_q + (1-\hat{q})\varepsilon_\alpha + \hat{q}\varepsilon_\beta + \varepsilon_c.$$

If simultaneous valid intervals are $q \in [q_L, q_U]$, $\alpha \in [a_L, a_U]$, $\beta \in [b_L, b_U]$, and $c \in [c_L, c_U]$, their exact rectangular drift bounds, provided the three probability intervals are contained in $[0, 1]$, are

$$L = (1 - q_U)a_L - q_U b_U - c_U, \qquad U = (1 - q_L)a_U - q_L b_L - c_L.$$

Proof. Subtract the true expression from the fitted expression as

$$\widehat{\mu} - \mu = (1 - \hat{q})(\widehat{\alpha} - \alpha) - \hat{q}(\widehat{\beta} - \beta) - (\hat{q} - q)(\alpha + \beta) - (\hat{c} - c).$$

Apply the triangle inequality and $\alpha + \beta \leq 2$. The drift is increasing in α and decreasing in q, β, c on their domains, so the stated corners attain its minimum and maximum on the rectangle. □

This propagation result assumes genuine interval coverage of the relevant conditional probabilities. It cannot manufacture such intervals from AUC, a chosen safety margin, cross-validation alone, or calibration plots.

2.7 What sequential uncertainty control can certify

Proposition 7 (a precisely scoped first-crossing guarantee). Suppose U_t is observable at decision t and

$$\mathbb{P}(\exists t \in \{m, \ldots, N-1\} : \mu_t > U_t) \leq \delta.$$

Let τ_U be the first $U_t \leq 0$ after the floor, with fallback N. Then

$$\mathbb{P}(\tau_U < N \text{ and } \mu_{\tau_U} > 0) \leq \delta.$$

Proof. On the simultaneous-coverage event, $\mu_{\tau_U} \leq U_{\tau_U} \leq 0$ whenever $\tau_U < N$. Every violation is therefore contained in the coverage-failure event. □

This certifies a nonpositive **one-step** drift, not optimality, retained accuracy, or proximity to a hindsight oracle. A corresponding simultaneous upper bound on Δ_t^* certifies the Bellman stop condition. To interpret a bound on μ_t as a statement about the myopic boundary one must separately impose the relevant structural assumptions.

For a probe construction, distinguish the pre-probe sigma-field $\mathcal{H}_t$ from the post-probe decision information $\mathcal{F}_t$. Let θ_t be an $\mathcal{H}_t$-measurable target, and assume the n_t probes $X_{t,i} \in [a, b]$ are conditionally independent given $\mathcal{H}_t$, each with conditional mean θ_t. Fix their count before observing them. Conditional Hoeffding gives a bound for θ_t, not automatically for μ_t:

$$U_t^{\mathrm{pre}} = \overline{X}_t + (b-a)\sqrt{\frac{\log(1/\delta_t)}{2n_t}}.$$

Taking $\delta_t = \delta/[r(r+1)]$ for decision index $r = t - m + 1 \geq 1$ gives $\sum_t \delta_t \leq \delta$, so a union bound establishes simultaneous coverage of the θ_t targets. Decisions can depend on earlier data; independence between different decision times is unnecessary. Within-prefix conditional independence and the correct conditional mean are essential. Adaptively choosing the number of probes requires a bound also uniform in probe count, or a further failure-budget allocation. The bounded-sample result is due to Hoeffding (1963).

Probe observations and their cost must enter the actual decision filtration. If they reveal information that changes the live conditional continuation gain, a bound for the pre-probe conditioning target is not automatically a bound for the post-probe target. Proposition 7 follows from this construction only if $\theta_t = \mu_t$ after the probes, or if an additional justified comparison supplies a bound for the post-probe μ_t. Shared unknown ground truth can make apparently independent generations dependent after

conditioning only on the visible prefix. Verified probe labels are also not generally available at deployment. An empirical-Bernstein formula requires its own theorem, variance convention, and sampling assumptions; merely inserting a sample variance does not prove those conditions. The modern confidence sequence framework is developed by Howard et al. (2021).

For completeness, a valid elementary e-process can be obtained when $X_i \in [-1, 1]$ is adapted to a sampling filtration $\mathcal{G}_i$ and the null satisfies $\mathbb{E}[X_i \mid \mathcal{G}_{i-1}] \geq 0$ for every i. For any fixed $\eta \in [0, 1)$,

$$E_n^{(\eta)} = \prod_{i=1}^{n} (1 - \eta X_i)$$

is a nonnegative supermartingale starting at 1: its next conditional multiplier has expectation at most 1. A fixed convex mixture of such processes has the same property. Finite optional sampling at the first threshold crossing, followed by monotone limits, gives

$$\mathbb{P}(\sup_n E_n \geq 1/\delta) \leq \delta.$$

This rejects that conditional-mean null for the specified sampling process. It does not estimate a different instance's present continuation gain. If $X_i = B$ for all i, with one shared fair sign B, all marginal means are zero, yet $(1 - X_i/2)$'s ten-factor product exceeds 20 on the negative branch with probability $1/2$. Hence marginal means and boundedness alone do not justify a 5% sequential guarantee.

In `trace_analysis.py`, empirical-Bernstein bounds and mixture products are computed across labeled task rows within a step, then used for a common step-level diagnostic. That construction is not an observable per-instance confidence bound. Repeated runs

of the same task also need dependence handling for population inference. This note makes no claim that the repository's raw fitted probabilities or historical detectors satisfy Proposition 7's coverage premise.

2.8 Scope of the empirical claims

Table 1. Mathematical objects and assumptions.

Mathematical object	Required information or assumptions	Defensible interpretation
Exact conditional hazard identity	Binary reward, correct conditioning, common transition sample	Algebraic one-step decomposition
Snell/Bellman policy	Conditional future law and all computation costs	Optimal finite-horizon causal stop
First nonpositive true drift	Persistent nonpositive drift after first crossing	Optimal under Theorem 4
Fitted first nonpositive drift	Causal features and validated probability estimates	Empirical myopic stopping rule
First nonpositive upper bound	Simultaneous coverage of the stated target	Probabilistic one-step sign certificate
Hindsight best recorded step	Completed labels and trace	Infeasible upper benchmark
Correctness AUC	Held-out target labels and scores	Ranking quality on that population

A hindsight maximum satisfies $\mathbb{E}[\max_{m\le t\le N}(C_t - K_t)] \ge \sup_\tau \mathbb{E}[C_\tau - K_\tau]$, because it dominates every realized admissible stop reward. Its maximizing time generally fails the stopping-time condition. Therefore a hindsight oracle gap includes unavailable information, and a reported early/late stop relative to that oracle is not a formal Type I error rate.

A minimum-step floor is a protocol constraint. It restricts the policy class and can improve a particular fitted policy empirically; it cannot improve the unconstrained optimum, because that optimum already includes every policy respecting the floor. For positive constant step cost, q_t measures accuracy while $q_t - \lambda(t - m)$ measures utility; their peaks need not coincide. Any savings claim must compare actual executed tokens/calls with a defined baseline, not convert stopped step indices into hardware savings without measurement.

The results are relative to the correctness target C_t. A passed grading test suite does not prove every empirical label correct, and an exact-answer benchmark theorem is not a theorem about open-ended semantic correctness. The standard optimal stopping machinery here is not claimed as a novel theorem or a universal law across model architectures.